Lab 4 AVCV 2026/2027: Fire-risk classification from satellite images

CUSTOM and FINE-TUNING networks on the FireRisk subset

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1. INTRODUCTION

Context and objective. [TODO: Task: predict one of 7 FireRisk classes from an RGB satellite image (image classification, not fire detection or forecasting). Motivation. Main challenges: class imbalance and visually similar, ordinal risk levels (Low/Moderate/High).]

Dataset. [TODO: FireRisk subset [1]: 5,861 RGB images of 320 × 320 px; 3,516 train / 1,172 val / 1,173 test (hidden labels). Stratified split: train counts 86/1,088/535/431/315/163/898 (Water, Very Low, Low, Moderate, High, Very High, Non-burnable), see Table 4. Network input: 224 × 224 crops (state any model at 320).]

Scope. [TODO: CUSTOM: VGG-style CNN from scratch, staged hyperparameter search. FINE-TUNING: ImageNet-pretrained torchvision models (ConvNeXt-T, Swin-T, ResNet-50). Main questions investigated (width/depth, epochs, class weights, augmentation, CutMix, ensembles, train+val).]

2. METHODOLOGY

2.1. Preprocessing and data augmentation

Pipeline. [TODO: RandomCrop 224 (train) / CenterCrop 224 (val, test) from the 320 image; float tensor; ImageNet mean/std normalisation. GPU batch augmentation: horizontal/vertical flips, 90° rotations, gamma, brightness, contrast, saturation (presets light/strong).]

Justification. [TODO: Satellite images have no canonical orientation; mild colour changes preserve vegetation cues. Validation evidence: strong vs light; mixup; CutMix (risk-only and standard) did not help; TTA lowered val F1; evaluating at 320 +0.002 (noise).]

2.2. System design: CUSTOM network

Architecture. [TODO: Final CUSTOM model and layer table (Table 5): 4 blocks of [Conv3x3-BN-ReLU]x2 + MaxPool, channels 48-96-192-384, global average pooling, FC 256 + ReLU + Dropout 0.3, FC 7. Params: 2.74 M. Kaiming init.]

Design decisions. [TODO: Differences with the reference (32-64-128-256? / Flatten / StepLR): GAP instead of Flatten (FC had 99% of the params), 2 convs per block (VGG [2]), BatchNorm [3], wider first block (+0.03 val F1), deterministic GAP.]

2.3. System design: FINE-TUNING network

Pretrained model. [TODO: Selected: Swin-T [4], torchvision `Swin_T_Weights.IMAGENET1K_V1` (ImageNet-1k [5]); head replaced by Linear(768, 7). Candidates: ConvNeXt-T [6], EfficientNetV2-S [7] (failed: non-finite loss with fp16), ResNet-50 [8]; AlexNet [9] as provided baseline.]

Adaptation strategy. [TODO: No frozen layers (full fine-tuning). ImageNet normalisation and 224 input. Discriminative LR: backbone lr, head lr = lr x head_mult (1 or 10) [10]; no weight decay on biases/norms. Stochastic depth [11]. Justify with validation (Table 1).]

2.4. Training protocol and model selection

Configuration. [TODO: Both models: cross-entropy with $1/\sqrt{\text{freq}}$ class weights, AdamW [12], linear warm-up + cosine decay [13], AMP [14]. CUSTOM: lr 3.17e-4, wd 0.0464, batch 128, 60 epochs. FT: lr 2.37e-5, wd 0.04, batch 32, 30 epochs. Seed 42. Kaggle Tesla T4, PyTorch [15] version X. No subsampling.]

Hyperparameter search. [TODO: Staged random search [16] with median-stopping pruning [17]: phases 1–2 (2 machines), phase 3 finalists, phases 4–5 targeted changes, deep search (121 trials); FT: 10 configs per network + 2 finalists per network at 30 epochs.]

Selection and reproducibility. [TODO: Checkpoint with the best validation macro-F1 (no early stopping in the final runs). Strict deterministic mode (`use_deterministic_algorithms`, fixed seeds, deterministic GAP): two identical 1-epoch runs give bit-identical weights. Train on train only; val only for selection.]

3. RESULTS AND ERROR ANALYSIS

Evaluation metrics. [TODO: Macro-F1 over the 7 labels; per-class precision/recall/F1 (Table 2); train vs val vs public.]

Experimental comparison. [TODO: Summary of Table 1: reference vs selected, under similar budgets; which changes helped (width 48x2, 60 epochs) and which did not (inv weights, 64x2, 5 blocks, CutMix, 80 epochs, ensembles, train+val).]

Learning curves and confusion matrices. [TODO: Figure 1: train/val loss and macro-F1. Figure 2: validation confusion matrices, row-normalised %, rows = true class.]

Visual error analysis. [TODO: Figure 3: validation images with true/predicted labels. Recurring confusions between adjacent risk levels (Low vs Very Low/Moderate). Train F1 in eval mode only 0.53 ⇒ limited visual signal. CUSTOM vs FT trade-offs.]

Competition feedback. [TODO: Public scores and submission IDs (Table 3); val-to-public gap of about 0.06 (selection on the same 1,172 validation images, noisy public subset).]

4. CONCLUSIONS

[TODO: Largest supported effects: CUSTOM (width/2 convs per block, GAP, 60 epochs); FT (pretraining itself, small backbone LR). Scratch vs fine-tuning. Limitations: small dataset, ordinal and ambiguous classes, validation reused for selection. Next steps: k-fold, ordinal losses, higher resolution, remote-sensing pretraining.]

Individual contributions. **Daniel Truzman:** [TODO: GPU data pipeline and AMP, phases 3–5 of the CUSTOM search (width 48x2, 60 epochs), strict determinism on Kaggle, CutMix / ensemble / train+val experiments, submissions.] **César Álvarez-Cascos:** [TODO: Parts 1–2 of the notebook, two-machine search (phases 1–2), deep hyperparameter search, fine-tuning search (Swin-T).]

Table 1: Main experiments (validation macro-F1). Reference and selected models in bold.

Cat.	Change	Objective	Budget	Val F1
CUSTOM	**Reference notebook**	Baseline	[TODO:]	[TODO:]
CUSTOM	Phases 1–2 random search	Architecture, optimiser, regularisation	15 ep	[TODO:]
CUSTOM	48 ch, 2 convs/block	More capacity (underfitting)	40 ep	0.496
CUSTOM	**60 epochs (n07__60ep)**	Longer schedule	60 ep	0.502
CUSTOM	[TODO: more rows]			
FT	AlexNet (provided)	Baseline	5 ep	[TODO:]
FT	**Swin-T (swin__p2__1)**	Selected FT	30 ep	0.546
FT	[TODO: more rows]			

Table 2: Per-class validation precision / recall / F1 of the selected models.

Model	Water	V. Low	Low	Mod.	High	V. High	Non-b.	Macro
CUSTOM F1	[TODO:]							
FT F1	[TODO:]							

Table 3: Public CodaBench scores (fixed undisclosed subset of the test set).

Submission ID	Date	Models (custom / ft)	CUSTOM	FT
1036062	05 Nov	p4__09__B / placeholder	0.49	–
[TODO:]				

[TODO: CUSTOM curves]

(a) CUSTOM

[TODO: FT curves]

(b) FINE-TUNING

Figure 1: Training/validation loss and macro-F1 per epoch.

[TODO: CUSTOM confusion matrix]

(a) CUSTOM

[TODO: FT confusion matrix]

(b) FINE-TUNING

Figure 2: Validation confusion matrices, row-normalised (%); rows = true class, columns = predicted class.

[TODO: grid of misclassified validation images (true / predicted)]

Figure 3: Representative validation errors (true → predicted).

Table 4: Class distribution.

Class	Train	Val
0 Water	86	[TODO:]
1 Very Low	1,088	
2 Low	535	
3 Moderate	431	
4 High	315	
5 Very High	163	
6 Non-burnable	898	

Table 5: Final CUSTOM architecture.

Stage	Layers	Output
Input	RGB crop	$3 \times 224 \times 224$
Block 1–4	[Conv3x3-BN-ReLU]×2, MaxPool	[TODO:]
Head	GAP, FC 256, ReLU, Dropout 0.3, FC 7	7

REFERENCES

[1] Shuchang Shen, Sachith Seneviratne, Xinye Wanyan, and Michael Kirley. FireRisk: A remote sensing dataset for fire risk assessment with benchmarks using supervised and self-supervised learning. In *International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, 2023.

[2] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations (ICLR)*, 2015.

[3] Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *International Conference on Machine Learning (ICML)*, 2015.

[4] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[5] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. ImageNet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015.

[6] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[7] Mingxing Tan and Quoc V. Le. EfficientNetV2: Smaller models and faster training. In *International Conference on Machine Learning (ICML)*, 2021.

[8] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[9] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2012.

[10] Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018.

[11] Gao Huang, Yu Sun, Zhuang Liu, Daniel Sedra, and Kilian Q. Weinberger. Deep networks with stochastic depth. In *European Conference on Computer Vision (ECCV)*, 2016.

[12] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[13] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations (ICLR)*, 2017.

[14] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, and Hao Wu. Mixed precision training. In *International Conference on Learning Representations (ICLR)*, 2018.

[15] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[16] James Bergstra and Yoshua Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13:281–305, 2012.

[17] Daniel Golovin, Benjamin Solnik, Subhodeep Moitra, Greg Kochanski, John Karro, and D. Sculley. Google vizier: A service for black-box optimization. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017.